

ALEKHYA THUMMALA

Senior Full Stack Java Developer | Java | Spring Boot | AWS | Microservices | AI-Enabled Applications

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PROFESSIONAL SUMMARY

- Senior Full Stack Java Developer with 10+ years of experience designing, developing, and deploying scalable enterprise applications using Java, Spring Boot, Angular, React, REST APIs, microservices, and cloud-native technologies.
- Strong expertise in developing enterprise-grade backend systems, reusable frontend components, secure API integrations, and distributed microservices architectures across financial services, healthcare, retail, telecom, and enterprise technology domains.
- Hands-on experience building scalable Java/J2EE applications using Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, RESTful Web Services, Microservices, and API-driven architecture.
- Experienced in developing responsive and interactive user interfaces using Angular, React.js, TypeScript, JavaScript ES6, HTML5, CSS3, Bootstrap, RxJS, Redux, and component-based frontend architecture.
- Strong experience implementing secure authentication and authorization mechanisms using Spring Security, OAuth2, JWT, API Gateway integrations, role-based access control, and secure service-to-service communication patterns.
- Expertise in designing event-driven systems and asynchronous processing workflows using Kafka, RabbitMQ, ActiveMQ, JMS, and real-time messaging patterns for high-volume distributed applications.
- Hands-on experience with AWS cloud services including EC2, S3, Lambda, RDS, IAM, CloudWatch, ECS, EKS, Auto Scaling, Elastic Load Balancing, and cloud-native deployment strategies.
- Proficient in containerization and DevOps practices using Docker, Kubernetes, Jenkins, Maven, Gradle, Git, CI/CD pipelines, automated deployment workflows, environment promotion, and release validation.
- Strong understanding of SQL and NoSQL databases including Oracle, PostgreSQL, SQL Server, MySQL, MongoDB, Cassandra, and DynamoDB with experience in query optimization, data modeling, and performance tuning.
- Experience developing and optimizing high-performance APIs and backend services supporting large-scale transactional systems, enterprise integrations, and high-volume data processing environments.
- Skilled in implementing automated testing frameworks and quality practices using JUnit, Mockito, Selenium, Cucumber, Jasmine, Karma, Postman, SOAP UI, integration testing, and regression testing methodologies.
- Experienced in production support, release validation, defect resolution, root cause analysis, centralized logging, application monitoring, and troubleshooting using Splunk, Kibana, Log4J, SLF4J, and CloudWatch.
- Exposure to AI-assisted application workflows, intelligent automation solutions, AI-enabled backend integrations, and modern data-driven engineering practices within enterprise cloud-native systems.
- Contributed to AI-friendly application capabilities by supporting scalable API services, intelligent data processing flows, workflow automation, recommendation-style logic, and data-driven application enhancements.
- Strong experience working in Agile/Scrum environments, collaborating with architects, QA teams, DevOps engineers, product owners, and business stakeholders to deliver scalable and reliable software solutions.
- Proven ability to translate complex business requirements into technical designs, reusable services, API contracts, database changes, production-ready features, and maintainable application architecture.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, TypeScript, Python, SQL, PL/SQL, Shell Scripting

Frontend Technologies: Angular, React.js, Redux, HTML5, CSS3, Bootstrap, Material UI, RxJS, AJAX, JSON

Backend Technologies: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, Microservices, Node.js, Express.js

Cloud & DevOps: AWS EC2, S3, Lambda, RDS, IAM, CloudWatch, ECS, EKS, Docker, Kubernetes, Jenkins, Maven,

Databases: Oracle, PostgreSQL, SQL Server, MySQL, MongoDB, Cassandra, DynamoDB, Stored Procedures, Query Optimization

Messaging & APIs: Kafka, RabbitMQ, ActiveMQ, JMS, REST, SOAP, Swagger/OpenAPI, OAuth2, JWT, API Gateway

Testing & Monitoring: JUnit, Mockito, Cucumber, Karma, Postman, SOAP UI, Splunk, Kibana, Log4J, SLF4J

Tools & Practices: Git, GitHub, Bitbucket, Jira, IntelliJ, Eclipse, Agile/Scrum, SDLC, TDD, Code Reviews

PROFESSIONAL EXPERIENCE

Freddie Mac | Sr. Full Stack Java Developer

Aug 2023 – Present

- Designed, developed, and enhanced enterprise-grade full stack applications using Java 17, Spring Boot, Angular 16, REST APIs, and microservices architecture to support business-critical mortgage technology workflows.
- Developed responsive and reusable Angular 16 components using TypeScript, HTML5, CSS3, Bootstrap, RxJS, routing, services, pipes, and reactive forms to deliver smoother user experience across application modules.
- Improved frontend performance by applying lazy loading, modular routing, reusable shared modules, optimized DOM rendering, and component-level code separation for scalable Angular applications.
- Implemented Spring Security, OAuth2/JWT-based authentication, role-based access checks, and secure API patterns to protect sensitive application workflows and user actions.
- Built scalable backend services using Spring Boot, Spring Data JPA, Hibernate, and REST APIs to improve service modularity, maintainability, and integration readiness.
- Integrated Kafka, JMS, and RabbitMQ-based asynchronous communication patterns to support event-driven processing, decoupled service interactions, and reliable message exchange across distributed systems.
- Containerized Java and Node-based application components using Docker and supported Kubernetes-based deployment workflows for scalable and consistent service delivery.
- Built and maintained Jenkins CI/CD pipelines for automated building, testing, code packaging, artifact deployment, and environment promotion, reducing manual release effort.
- Reduced API response time by approximately 25% by optimizing backend service logic, SQL queries, repository methods, and API response handling.
- Improved UI performance by approximately 30% through Angular lazy loading, modular architecture, optimized frontend rendering, and shared component reuse.
- Enhanced system reliability by improving centralized logging, monitoring, alerting, and proactive issue detection using Splunk, Kibana, Log4J, SLF4J, and CloudWatch-style observability practices.
- Contributed to AI-assisted workflow enhancement initiatives by supporting intelligent data processing flows, backend integrations, and scalable API services for enterprise applications.
- Worked on integrating AI-enabled and data-driven features into microservices architecture to improve operational efficiency, workflow automation, user experience, and data-backed decision support.
- Supported AI-friendly application patterns by preparing backend APIs, structured datasets, data validation workflows, and service endpoints used by intelligent automation and analytics features.
- Developed unit and integration test cases using JUnit and Mockito to validate business logic, API behavior, service-layer functionality, and release readiness before deployment.
- Collaborated with product owners, QA teams, architects, and DevOps engineers in Agile ceremonies, sprint planning, backlog refinement, code reviews, release coordination, and production support.
- Supported production deployments, defect triage, root cause analysis, hotfix validation, and release verification while maintaining clean coding standards and reusable design patterns.
- Improved deployment efficiency and reduced release timelines by streamlining Jenkins pipelines, automating deployment steps, and improving environment validation workflows.

Cardinal Health | Sr. Java Full Stack Developer

Apr 2022 - Jul 2023

- Designed and implemented enterprise full stack applications using Java, Spring Boot, Angular 11, TypeScript, REST APIs, and microservices architecture in Agile delivery environments.
- Developed reusable Angular components, services, mappers, constants, form validations, and UI modules to improve application maintainability and frontend development efficiency.
- Built REST-based microservices using Spring Boot, Spring Data JPA, Spring MVC, Spring Security, Hibernate, DTOs, repositories, and layered backend architecture.
- Implemented secure API integrations using OAuth/JWT-based authentication, API Gateway patterns, role-based authorization flows, and secure request validation.
- Developed Kafka producer and consumer workflows for asynchronous event processing, improving system scalability, responsiveness, and reduced service coupling.

- Worked with AWS services including EC2, S3, Lambda, ECS, RDS, IAM, CloudWatch, and Elastic Load Balancing for scalable application hosting and monitoring.
- Created and improved CI/CD pipelines using Jenkins, Maven, Git, Docker, Kubernetes, and automated deployment scripts to streamline release workflows.
- Optimized SQL queries, repository methods, and backend service logic across Oracle, SQL Server, MySQL, and MongoDB environments to improve data access performance.
- Developed automated regression and integration tests using JUnit, Mockito, Selenium WebDriver, and Cucumber to improve test coverage and release confidence.
- Used Postman, Swagger/OpenAPI, SOAP UI, Splunk, Kibana, and Log4J for API validation, debugging, monitoring, and troubleshooting service behavior.
- Collaborated with QA teams, DevOps engineers, architects, product owners, and business stakeholders to support feature delivery, defect resolution, and release readiness.
- Supported production deployments, environment validation, release verification, defect fixes, and application support activities across multiple modules.
- Worked on backend data validation, response transformation, and API mapping logic to support consistent integrations between frontend modules and backend services.
- Participated in Agile ceremonies, technical discussions, code reviews, sprint planning, and retrospective meetings to improve delivery quality and team collaboration.

Merck Pharma | Full Stack Developer

May 2020 - Mar 2022

- Developed enterprise healthcare and pharmaceutical applications using Java 8/11, Spring Boot, Angular 8, REST APIs, Hibernate, Node.js, and AWS cloud technologies.
- Built dynamic frontend screens using Angular, HTML5, CSS3, Bootstrap, JavaScript, JSON, AJAX, and reusable component-based design patterns.
- Designed and implemented backend APIs using Spring Boot, Spring JPA, Hibernate, Swagger/OpenAPI, RESTful service architecture, and layered service components.
- Integrated RabbitMQ and Kafka messaging systems to support asynchronous communication, event-based processing, and reliable data exchange between services.
- Implemented logging and troubleshooting mechanisms using Log4J, SLF4J, Kibana, and Splunk for runtime issue analysis and production support.
- Optimized SQL Server and PostgreSQL queries, stored procedures, database scripts, and search filters to support reporting and transactional workflows.
- Worked with AWS services including EC2, S3, RDS, SQS, IAM, CloudWatch, EBS, and CloudFront for cloud-based application deployment and monitoring.
- Developed unit and integration test cases using JUnit and Mockito and supported automated regression validation through Jenkins pipelines.
- Implemented Hibernate ORM mappings, DAO layers, repository components, and backend persistence logic to support reliable data access.
- Collaborated with QA teams and business stakeholders to analyze requirements, validate functionality, resolve defects, and deliver enterprise application features.
- Supported production systems, defect triage, debugging, deployment validation, and release coordination across multiple application modules.
- Improved application stability by supporting centralized logging, structured exception handling, database validation, and backend service monitoring.
- Worked on secure API interactions, payload validation, service-layer changes, and database updates to support internal business workflows.
- Participated in Agile ceremonies, sprint discussions, release planning, and code reviews to ensure timely delivery of high-quality software
- Worked with Spring Security, Spring Cloud, REST services, and service discovery concepts to support microservices-based application development.
- Performed application troubleshooting by reviewing logs, validating APIs, analyzing database issues, and coordinating fixes during release cycles.

PNC Bank | Full Stack Developer

Mar 2018 - Apr 2020

- Developed enterprise banking applications using Java, Spring Boot, Spring Security, Angular 8, REST APIs, MongoDB, SQL Server, Kafka, and AWS cloud services.
- Designed responsive UI screens using HTML5, CSS3, JavaScript, Bootstrap, jQuery, AJAX, and Angular component-based architecture for banking workflows.
- Built REST APIs using Spring Boot and integrated frontend modules with backend services using structured JSON request/response patterns.
- Implemented Kafka-based streaming workflows for real-time data movement, asynchronous processing, and distributed service communication.
- Worked with MongoDB, SQL Server, MySQL, and Oracle for backend persistence, query development, stored procedures, and data validation logic.
- Created and configured AWS EC2 instances, S3 buckets, Lambda functions, and cloud-based application resources for scalable deployments.
- Developed automation and regression test cases using Selenium WebDriver, JUnit, Cucumber, and Jenkins-based CI/CD workflows.
- Used Spring Security, Hibernate, JPA, and service-layer components to build secure backend features and reusable business logic.
- Enhanced application quality by supporting test-driven development, debugging, defect resolution, and production release validation.
- Participated in Agile ceremonies including sprint planning, code reviews, backlog discussions, defect triage, and release coordination.
- Improved application maintainability by building reusable components, standardizing API patterns, and supporting modular design practices.
- Supported production issue analysis by reviewing logs, validating database records, testing API responses, and coordinating fixes with QA teams.
- Worked with Docker, Jenkins, Git, Maven, and deployment scripts to support automated builds and environment promotion activities.
- Collaborated with cross-functional teams to deliver banking application enhancements while maintaining secure coding and performance standards.

Verizon | Full Stack Developer

Jul 2016 - Dec 2017

- Developed enterprise telecom applications using Java, Spring Boot, Spring Cloud, React.js, Redux, REST APIs, Kafka, and microservices architecture.
- Built single-page applications using React, Redux, React Router, ES6, Bootstrap, and reusable frontend components for responsive user interfaces.
- Developed Spring Boot microservices and RESTful APIs to support distributed application workflows and backend service integrations.
- Implemented Kafka-based streaming and messaging workflows for real-time data processing, asynchronous communication, and service integration.
- Created Splunk dashboards for application monitoring, performance tracking, log analysis, and production issue troubleshooting.
- Deployed containerized applications using Docker and Kubernetes with AWS-based infrastructure support for scalable application hosting.
- Developed unit, integration, UI, and performance test cases using JUnit, Karma, Protractor, JMeter, SOAP UI, and Jenkins pipelines.
- Supported database development using SQL Server, MySQL, Oracle, Hibernate, PL/SQL packages, stored procedures, and query optimization techniques.
- Participated in release management, deployment validation, debugging, defect resolution, and production support activities.
- Collaborated with development, QA, and operations teams to enhance application scalability, maintainability, and service reliability.
- Supported build and deployment automation using Jenkins, GitHub, Maven, Ant, and environment-specific configuration updates.

Dhruvsoft Services | Java Developer

Jul 2015 - Jun 2016

- Developed Java-based web applications using Core Java, Spring, Hibernate, JSP, Servlets, JavaScript, jQuery, HTML, CSS, and SQL.
- Built service-layer components using Spring IOC, dependency injection, REST/SOAP services, and microservices-based design practices.
- Developed UI screens and backend business logic using Java, Spring, Hibernate, PHP, XML, XSLT, and database-driven workflows.
- Worked with Oracle database by creating tables, SQL queries, PL/SQL procedures, triggers, database scripts, and validation logic.
- Developed JUnit test cases and used Log4J for application logging, runtime issue tracking, and debugging.
- Used Kafka/message queues for asynchronous communication between enterprise business applications and backend systems.
- Participated in Agile ceremonies, requirement gathering, analysis, design discussions, development, unit testing, and deployment support.
- Created shell scripts to monitor application status, support production operations, and automate routine application tasks.
- Implemented Hibernate persistence logic and DAO components to support database updates and application transactions.
- Collaborated with senior developers and QA teams to resolve defects, validate functionality, and support release activities.
- Supported XML/XSLT-based data transformation workflows for payment, transaction, and reporting-related application modules.
- Assisted in performance tuning, debugging, reusable code development, and standardized coding practices across application components.